CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 12: **Tales by Dots and Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mean of 5 observations 16, 30, 17, 32, 25 is 24. If a new observation 36 is added, the new mean is: 1

(a) 26 **(b)** 24 **(c)** 60 **(d)** 27

2. A plant's height (cm) at the end of weeks 1 to 6 is 12, 12, 16, 21, 21, 19. Between which two consecutive weeks was the growth the greatest? 1

(a) Between week 2 and week 3 **(b)** Between week 3 and week 4
(c) Between week 1 and week 2 **(d)** Between week 4 and week 5

3. A dot plot shows: 9 dots at 3, 6 dots at 4, 7 dots at 5, 1 dots at 6, 8 dots at 7, 5 dots at 8. The mode of the data is: 1

(a) 6 **(b)** 4 **(c)** 3 **(d)** 5

4. The values 3, 7, 9, 11 occur 1, 1, 6, 4 times respectively. The mean is: 1

(a) 7.5 **(b)** 3 **(c)** 30 **(d)** 9

5. The median of 30, 10, 42, 18, 13, 31, 35, 26 is: 1

(a) 25.62 **(b)** 30 **(c)** 28 **(d)** 26

6. The mean of 5 numbers is 50. If 4 of them are 45, 55, 60, 38, the remaining number is: 1

(a) 52 **(b)** 49 **(c)** 50 **(d)** 57

7. The mean of 5 observations 24, 18, 33, 22, 23 is 24. If a new observation 12 is added, the new mean is: 1

(a) 24 **(b)** 27 **(c)** 36 **(d)** 22

8. A plant's height (cm) at the end of weeks 1 to 6 is 8, 5, 11, 14, 19, 23. Between which two consecutive weeks was the growth the greatest? 1

(a) Between week 4 and week 5 **(b)** Between week 2 and week 3
(c) Between week 3 and week 4 **(d)** Between week 1 and week 2

9. A dot plot shows: 7 dots at 3, 7 dots at 4, 8 dots at 5, 2 dots at 6, 6 dots at 7, 2 dots at 8. The mode of the data is: 1

(a) 5 **(b)** 6 **(c)** 4 **(d)** 3

10. The values 6, 7, 9, 10 occur 1, 2, 2, 1 times respectively. The mean is: 1

(a) 1.5 **(b)** 9 **(c)** 32 **(d)** 8

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Each of the data values 6, 7, 25, 18, 19 is increased by 10. Find the old and the new mean. 2

12. Find the median of 46, 57, 32, 54, 24, 16. 2

13. The mean of 4 numbers is 52. Three of them are 48, 60, 45. Find the fourth. 2

14. Each of the data values 7, 22, 10, 28, 23 is increased by 3. Find the old and the new mean. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The runs scored by a batter in 6 innings are 27, 58, 29, 40, 57, 11. Find the mean, the median and the range. 3

16. The marks 3, 6, 9, 12 were obtained by 2, 4, 3, 1 students respectively. Find the mean mark. 3

17. The mean of 5 observations is 15. One observation, 15, is removed. Find the new mean. 3

OR

The runs scored by a batter in 6 innings are 21, 60, 45, 86, 88, 24. Find the mean, the median and the range. 3

18. The marks 50, 60, 70, 80 were obtained by 4, 2, 3, 1 students respectively. Find the mean mark. 3

OR

The mean of 7 observations is 36. One observation, 36, is removed. Find the new mean. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The monthly rainfall (mm) from June to November is 174, 101, 152, 121, 195, 133. 5
(i) Find the mean rainfall.
(ii) Which month had the greatest rise over the previous month?
(iii) Find the range.

OR

The runs of two players in 5 matches are: A: 36, 61, 79, 67, 17; B: 66, 78, 54, 67, 30. Find the mean of each and say who is more consistent (smaller range). 5

20. The monthly rainfall (mm) from June to November is 66, 167, 165, 120, 181, 189. 5
(i) Find the mean rainfall.
(ii) Which month had the greatest rise over the previous month?
(iii) Find the range.

OR

The runs of two players in 5 matches are: A: 28, 43, 22, 74, 13; B: 11, 66, 86, 46, 26. Find the mean of each and say who is more consistent (smaller range). 5

END OF QUESTION PAPER — Total: 40 Marks